

2022

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer any five questions in which

Q. No.1 is compulsory.

1. Answer any four of the following : $3\frac{1}{2} \times 4 = 14$
 - (a) Use Maxwell's equations and obtain wave equation for free space.
 - (b) Find the Lorentz transformation of x-component of the vector potential.
 - (c) Explain the meaning of Non-uniqueness of electromagnetic potentials.
 - (d) Define retarded time and retarded potentials.

- (e) Define Scalar and Vector potentials.
 - (f) Define Plasma. Mention its general properties.
 - (g) What do you mean by plasma confinement ?
 - (h) What is Pinch effect ?
2. Obtain four vector form of four vector potential (A^μ) and the four current density vector (j^μ). 7+7 = 14
 3. Define electromagnetic field tensor ($F^{\mu\nu}$). Use $\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$ and obtain corresponding two Maxwell's equations. 6+8 = 14
 4. Write down the expression of Lienard-Wiechert potentials and obtain an expression of electric field due to a moving point charge. 4+10 = 14
 5. What do you mean by radiation damping ? Obtain Abraham-Lorentz formula. 5+9 = 14
 6. Explain and deduce Magneto Hydrodynamics (MHD) equations for plasma. 14
 7. What is Pinch effect ? Explain its mechanism. 14

8. Discuss, in detail, the theory of Faraday rotation of plasma. 14

9. Write short notes on any two of the following :

7×2 = 14

- (a) Dual field tensor
- (b) Coulomb and Lorentz gauge
- (c) Debye-Shielding length
- (d) Kinetic theory of plasma

